

Multi-Agent Systems: Homework Assignment 1

Alexandru Licuriceanu
alicuriceanu@stud.acs.upb.ro

1 Environments

This assignment evaluates the performance of three reinforcement learning algorithms and how their hyperparameters influence the results. For this purpose, we will implement and test SARSA, Q-learning, and Double Q-learning in navigating two distinct grid-based maze environments, seen in Figure 1. For the multi-agent task, the first agent starts from position (2, 0), and the second agent from position (4, 0).

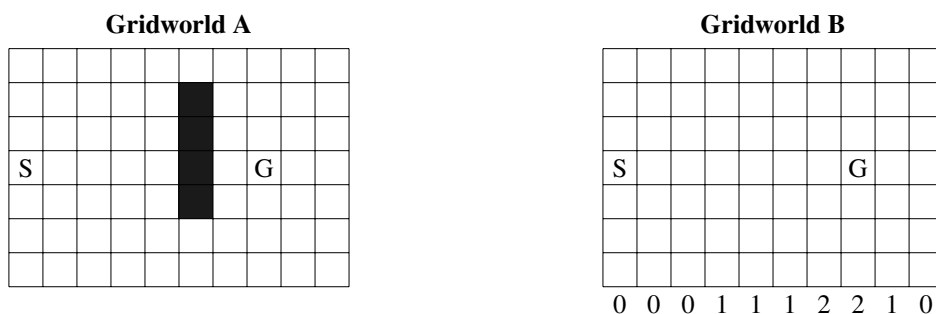


Figure 1: Gridworld environments: Gridworld A has static obstacles; Gridworld B includes a crosswind.

1.1 Gridworld A

- Layout: A standard grid where the agent can move up, down, left, and right.
- Challenge: The environment contains obstacles (black cells) that block the agent's path to the goal.

1.2 Gridworld B

- Layout: The same baseline grid, but without the static obstacles.
- Challenge: A crosswind blows upward through the columns of the grid. When the agent moves, its resulting state is shifted upward by the specific column's wind strength.

2 Single-Agent Task 1

To evaluate the performance of SARSA, Q-Learning, and Double Q-Learning, I performed ablation studies across varying learning rates ($\alpha \in \{0.1, 0.5, 0.9\}$) and exploration rates ($\epsilon \in \{0.01, 0.1, 0.4\}$). The base parameter values used when the other was varied are: $\alpha = 0.5$, $\epsilon = 0.1$. All experiments were run 10 times each.

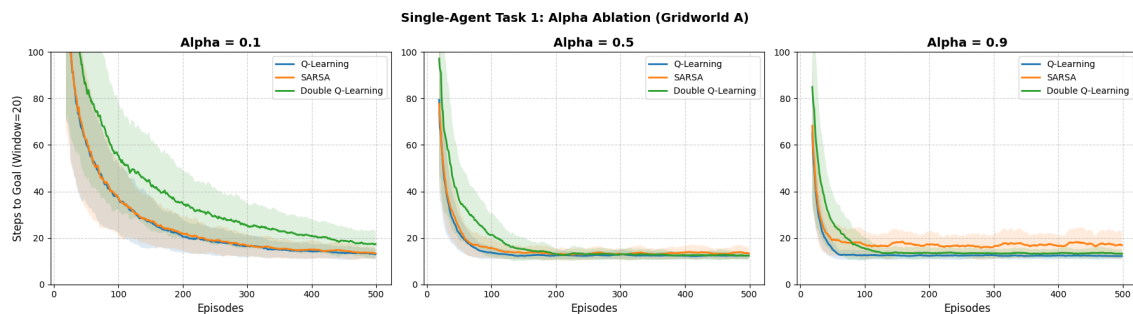


Figure 2: Alpha hyperparameter ablation for Gridworld A.

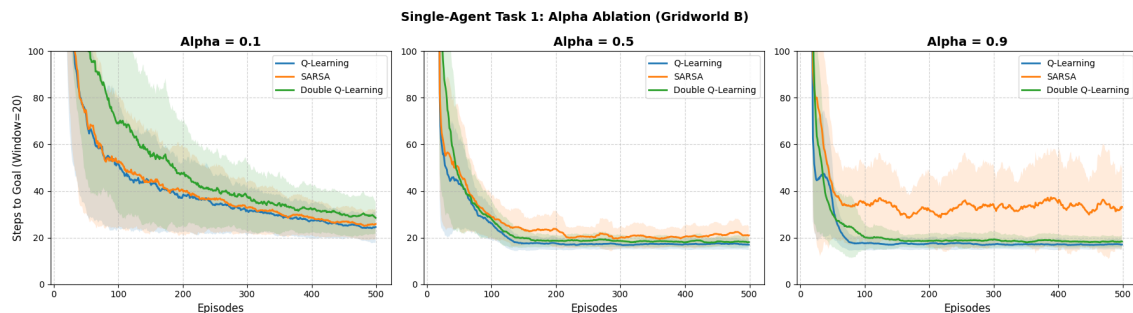


Figure 3: Alpha hyperparameter ablation for Gridworld B.

2.1 Convergence Time across α Variations

Gridworld A Figure 2

- Low Learning Rate ($\alpha = 0.1$): Convergence is slow, requiring roughly 300 to 400 episodes across all algorithms to fully flatten for SARSA and Q-Learning and around 500 for Double Q-Learning.
- Medium Learning Rate ($\alpha = 0.5$): All algorithms converge correctly at a perfect rate to the optimal paths, at ~ 200 episodes.
- High Learning Rate ($\alpha = 0.9$): Convergence accelerates dramatically. At $\alpha = 0.9$, Q-Learning and Double Q-Learning find the optimal path in under 100 episodes. For $\alpha = 0.9$, SARSA maintains a higher floor and less stable results compared to Q-Learning and Double Q-Learning.

- Because the environment lacks stochasticity, heavily weighting new information ($\alpha = 0.9$) allows the agents to rapidly map the perimeter of the obstacles without destabilizing their established policies.

Gridworld B Figure 3

- Low Learning Rate ($\alpha = 0.1$): Fails to achieve convergence to any optimal path for any algorithm within the allotted episodes.
- Medium Learning Rate: Again, all algorithms are balanced and converge to an optimal path, although SARSA takes slightly more steps to reach the goal in its final policy.
- High Learning Rate ($\alpha = 0.9$) for Off-Policy Agents: Q-Learning and Double Q-Learning successfully handle the aggressive rate, reducing their convergence time to approximately 100 episodes.
- High Learning Rate ($\alpha = 0.9$) for SARSA: Exhibits a complete failure to converge. The learning curve oscillates heavily and never flattens into a stable, near-optimal path.
- Because SARSA updates its Q-values based on the action it actually executes next, an aggressive learning rate causes it to overwrite and unlearn its value estimates every time it is heavily penalized by a wind shift, making it unable to converge to the optimal value.

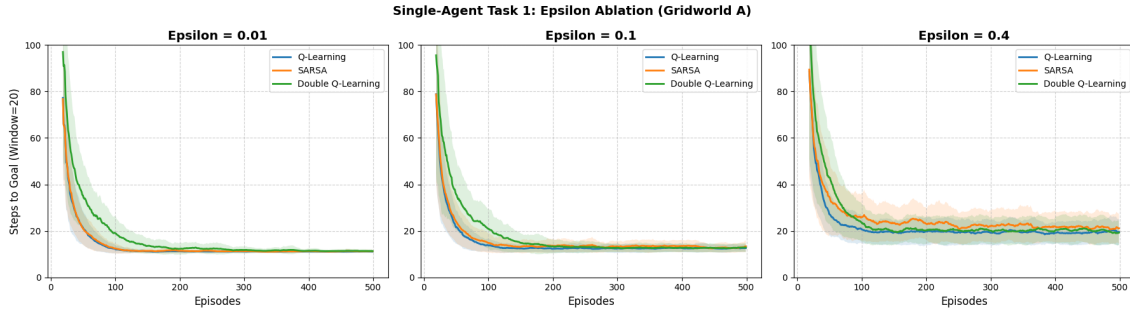


Figure 4: Epsilon hyperparameter ablation for Gridworld A.

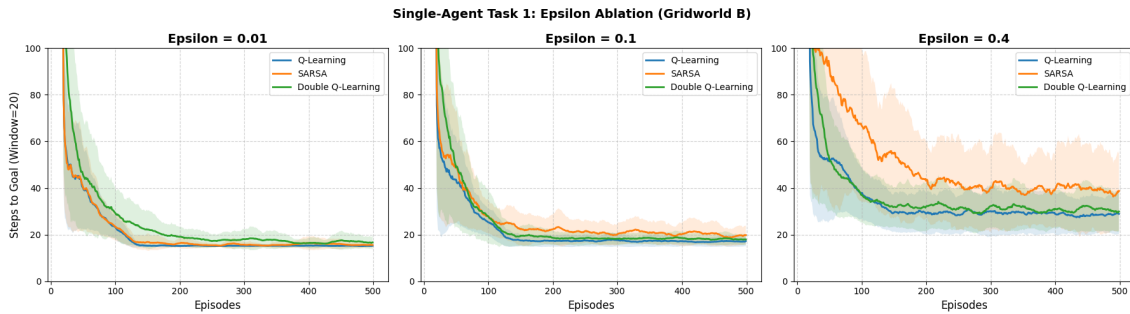


Figure 5: Epsilon hyperparameter ablation for Gridworld B.

2.2 Convergence Time across ϵ Variations

Gridworld A Figure 4

- Low Exploration ($\epsilon \in \{0.01, 0.1\}$): All three algorithms yield near-identical, highly efficient paths, converging to a baseline of roughly 14 steps.
- High Exploration ($\epsilon = 0.4$) General Impact: Path efficiency inherently degrades for all agents, as they are mathematically forced to execute random moves 40% of the time.
- High Exploration for Off-Policy Agents: Q-Learning and Double Q-Learning maintain a relatively tight efficiency floor (~ 20 steps). Their update rules ignore forced randomness, allowing them to maintain the theoretically shortest path.
- High Exploration for SARSA: Settles at a slightly higher step count (~ 23 steps) with wider variance.

Gridworld B Figure 5

- Low Exploration ($\epsilon \in \{0.01, 0.1\}$): All algorithms converge well, though SARSA consistently takes a slightly longer route at $\epsilon = 0.1$.
- High Exploration ($\epsilon = 0.4$) General Impact: The environmental penalty for random moves becomes severe; straying randomly in the grid's center subjects the agent to massive upward displacement.
- High Exploration for Off-Policy Agents: Q-Learning and Double Q-Learning prove highly robust. Despite being blown off course constantly, their target policies remain stable, allowing the learning curve to flatten around 30 steps.
- High Exploration for SARSA: Experiences a complete breakdown. The learning curve shoots to an average of 40-50 steps with erratic variance bands. Because it updates based on actual next actions, the frequent random penalties cause SARSA to become excessively paranoid of the wind, abandoning good paths and failing to converge.

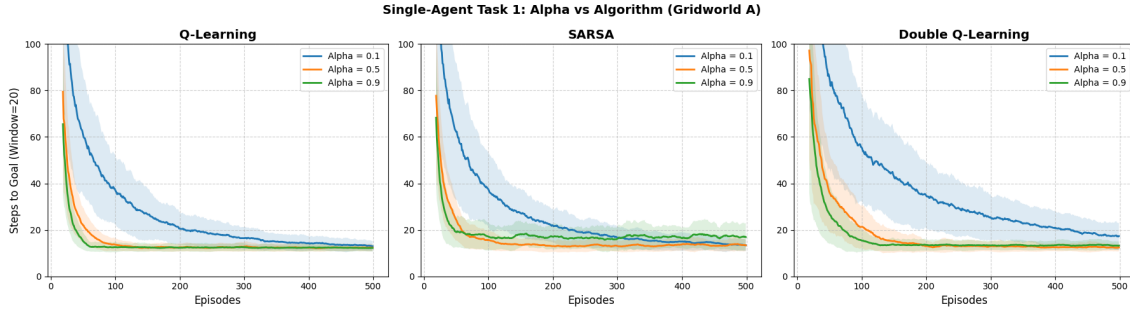


Figure 6: Alpha hyperparameter comparison, per algorithm, for Gridworld A.

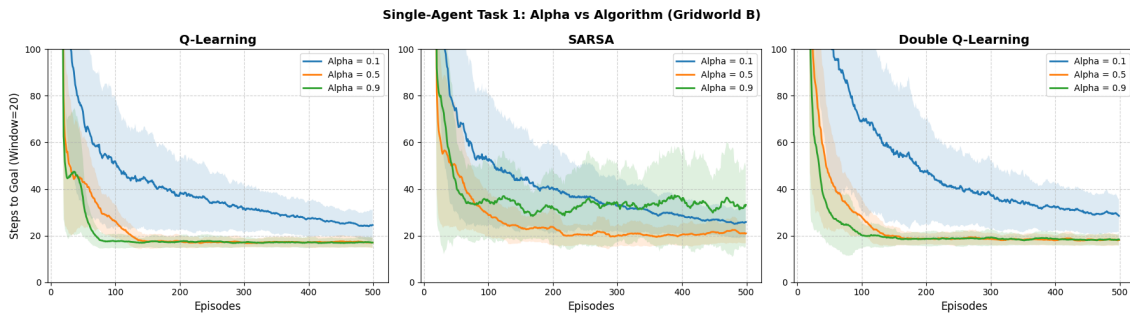


Figure 7: Alpha hyperparameter comparison, per algorithm, for Gridworld B.

2.3 Robustness to (α) Variations

Gridworld A Figure 6

- As the environment is fully deterministic, all three algorithms exhibit nearly identical, highly robust behavior across the α variations.
- Low Learning Rate ($\alpha = 0.1$, blue curves): Convergence is smooth but slow, requiring over 300 episodes to flatten for Q-Learning and SARSA, and 500 episodes for Double Q-Learning.
- High Learning Rate ($\alpha = 0.5$ and $\alpha = 0.9$, orange and green curves): Convergence time accelerates drastically to under 100 episodes, maintaining excellent final path efficiency without introducing instability.
- Because Gridworld A lacks environmental noise, aggressive updates ($\alpha = 0.9$) are beneficial. The agents can immediately trust the discovered rewards without needing to compensate for stochastic anomalies, resulting in fast, stable convergence across all algorithms.

Gridworld B Figure 7

- Q-Learning & Double Q-Learning): Both remain highly robust, converging to an efficient path even at the highest learning rate ($\alpha = 0.9$). Because their updates use the greedy maximum of the next state, they evaluate the optimal path rather than the exploratory path, effectively ignoring the massive penalties incurred when occasional ϵ -driven random moves force them into disadvantageous wind shifts.
- Double Q-Learning appears slightly more stable than standard Q-Learning, exhibiting tighter variance bands by successfully mitigating maximization bias. Neither Q-Learning or Double Q-Learning converge to the optimal path when using low learning rates ($\alpha = 0.1$).
- While SARSA converges smoothly at lower learning rates ($\alpha = 0.1$ and $\alpha = 0.5$), it suffers a complete breakdown at $\alpha = 0.9$.
- This instability stems from SARSA's on-policy nature: it evaluates the action it actually executes. When combined with an aggressive learning rate ($\alpha = 0.9$), SARSA overreacts to its own ϵ -driven exploratory mistakes. When the wind sweeps the agent away, it severely downgrades the Q-values of the safe states that preceded the random move, effectively unlearning its optimal path.

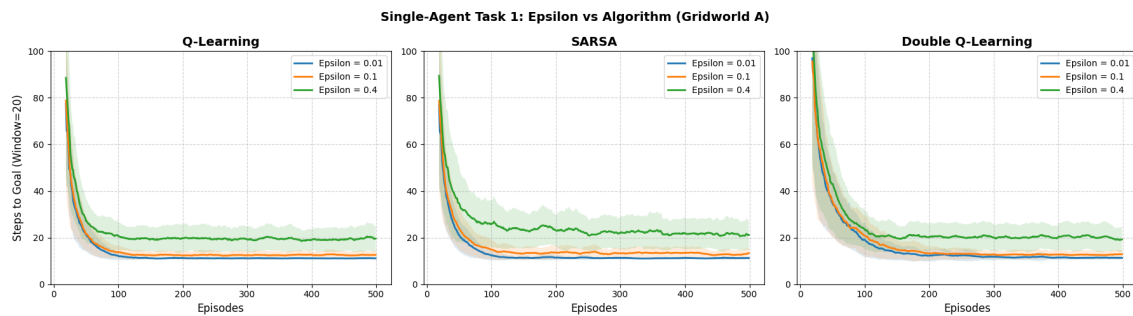


Figure 8: Epsilon hyperparameter comparison, per algorithm, for Gridworld A.

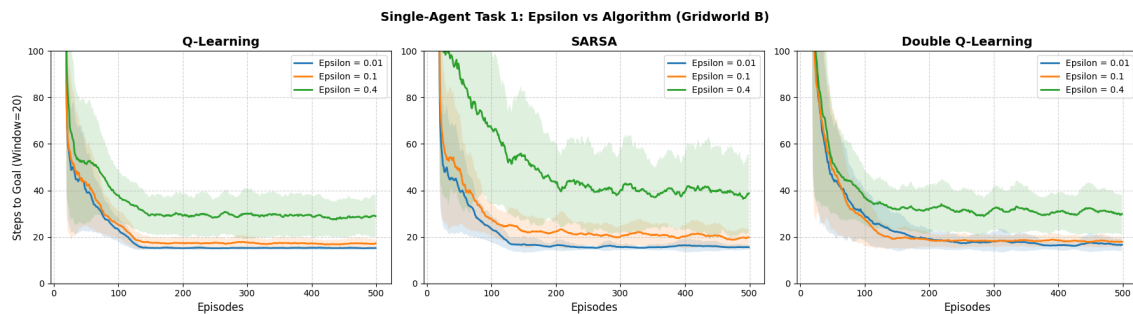


Figure 9: Epsilon hyperparameter comparison, per algorithm, for Gridworld B.

2.4 Robustness to (ϵ) Variations

Gridworld A Figure 8

- All three algorithms handle the environment well. At $\epsilon = 0.01$ and $\epsilon = 0.1$, they quickly converge to the optimal path.
- For high Exploration rates ($\epsilon = 0.4$), the step count naturally shifts upward for all algorithms, and do not converge to an optimal or near-optimal path, as they're taking random actions 40% of the time.

Gridworld B Figure 9

- Q-Learning & Double Q-Learning: Even at $\epsilon = 0.4$, these algorithms stabilize at a high plateau of ~ 30 steps. Because their updates ignore the behavior policy ($\max_a Q(s', a)$), their underlying Q-values still learn the optimal route. This allows the curve to flatten smoothly, even though the pathfinding is again influenced by 40% randomness.
- SARSA stabilizes even higher for $\epsilon = 0.4$ at ~ 40 steps.

2.5 Takeaways & Insights

- The absence or presence of environmental noise fundamentally dictates how aggressively an agent can learn. In Gridworld A, aggressive updates do not destabilize the agents, whereas in Gridworld B, environmental penalties expose the structural vulnerabilities of on-policy learning.
- Aggression is rewarded in static environments. In Gridworld A, a high learning rate ($\alpha = 0.9$) is highly beneficial, drastically reducing convergence time to under 100 episodes for Q-learning and Double Q-learning.
- In Gridworld B, Q-Learning and Double Q-Learning handle aggressive learning rates ($\alpha = 0.9$) successfully. Because they update based on the greedy maximum of the next state, they evaluate the optimal path and effectively ignore the massive penalties incurred when occasional random moves shift them into the wind.
- SARSA suffers a complete breakdown and fails to converge at $\alpha = 0.9$ in Gridworld B. Because SARSA updates based on the action it actually executes, it "overreacts" to its own exploratory mistakes.
- High exploration guarantees sub-optimal execution: When exploration is pushed to $\epsilon = 0.4$, path efficiency inherently degrades for all agents across all environments, as they are mathematically forced to take random actions 40% of the time.
- SARSA learns safer but longer paths. In Gridworld A under high exploration, SARSA anticipates that it will frequently stumble into obstacles. To compensate, it artificially downgrades states near those obstacles, settling on a safer but noticeably longer trajectory (~ 25 steps) compared to the off-policy agents.
- In both environments, Q-Learning and Double Q-Learning maintain a tighter, more efficient floor under high exploration. Their target policies remain stable because their update rules ignore the random behavior policy.

- SARSA’s paranoia in punishing environments: In Gridworld B, $\epsilon = 0.4$ subjects the agent to massive upward wind displacement. This causes SARSA to experience a complete breakdown, with step counts oscillating wildly between 40 and 60. The frequent random penalties create a negative feedback loop, making SARSA excessively “paranoid” of the wind, forcing it to abandon good paths and preventing convergence.

3 Single-Agent Task 2

To evaluate the impact of a richer action space, we re-solved Gridworld B using SARSA with eight possible actions (including diagonals) and compared it to the baseline four-action SARSA from Task 1. As seen in Figure 10, the introduction of diagonal moves dramatically transforms the agent’s capabilities:

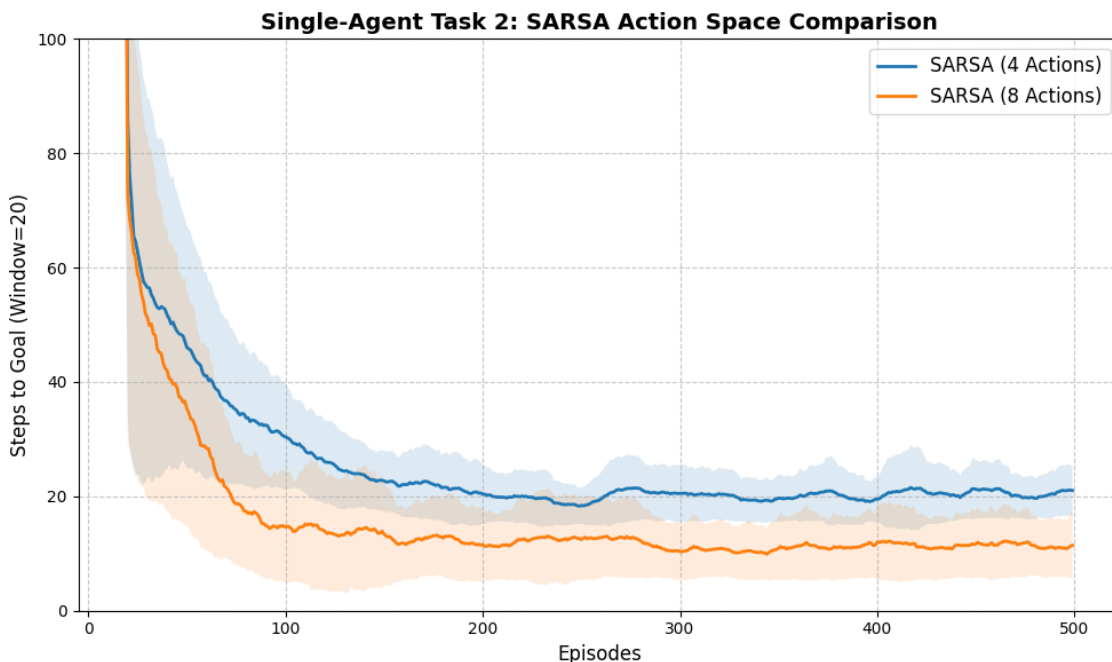


Figure 10: SARSA Comparison on Gridworld B (4 moves vs. 8 moves).

- Path Efficiency: The biggest difference is the reduction in steps required to reach the goal, which comes as no surprise. The 8-action SARSA (orange curve) converges to a highly efficient floor of approximately 10 to 12 steps. In contrast, the 4-action SARSA (blue curve) bottoms out at roughly 20 to 22 steps. This represents a near 50% improvement in path efficiency.
- The Mechanics: For example, in a 4-action space, if the agent is pushed upward by the wind, it must waste an entire step moving down to correct its row before it can move Right toward the goal. With an 8-action space, the agent can choose to move down and right in a single move. This single diagonal

action simultaneously counteracts the vertical wind displacement and advances the agent horizontally, effectively halving the steps needed to traverse the highest wind columns.

- **Faster Convergence:** The 8-action SARSA curve flattens noticeably earlier, finding its optimal, stable path in under 100 episodes. The 4-action variant struggles longer against the wind, requiring closer to 150-200 episodes to fully stabilize.
- **Tighter Variance and Robustness:** The variance band for the 8-action agent is tighter after convergence. Because the agent has more granular, directional control to fight the wind, occasional ϵ -driven random exploratory moves are easier to recover from, making the final policy much more robust to ongoing exploration.

4 Multi-Agent Task 1

In order for the algorithms to actually converge, compared to the single-agent case, we have increased the number of episodes from 500 to 5000. We used two agents for this task, starting from positions (2, 0) and (4, 0). All experiments were run 10 times each.

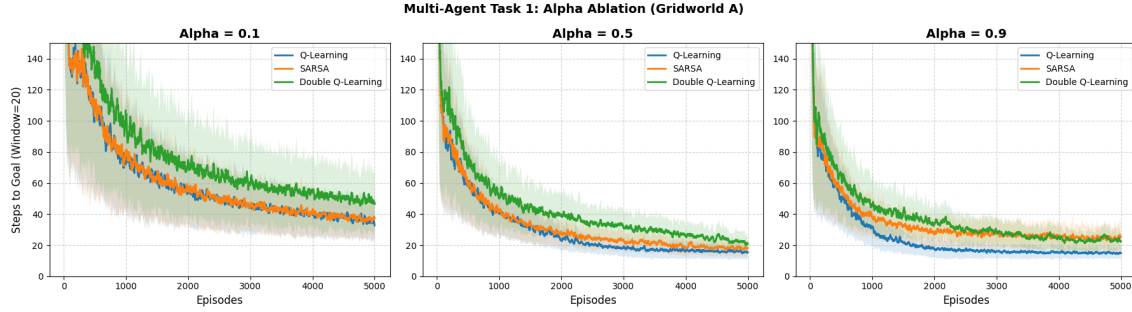


Figure 11: Alpha hyperparameter ablation for Gridworld A, multi-agent case.

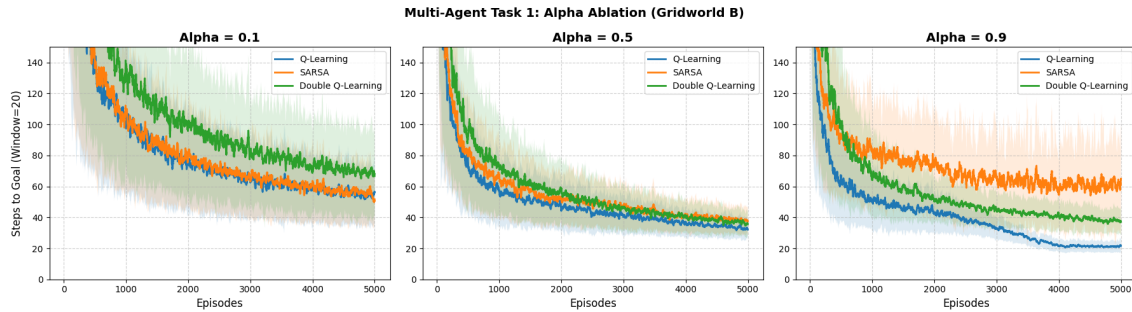


Figure 12: Alpha hyperparameter ablation for Gridworld B, multi-agent case.

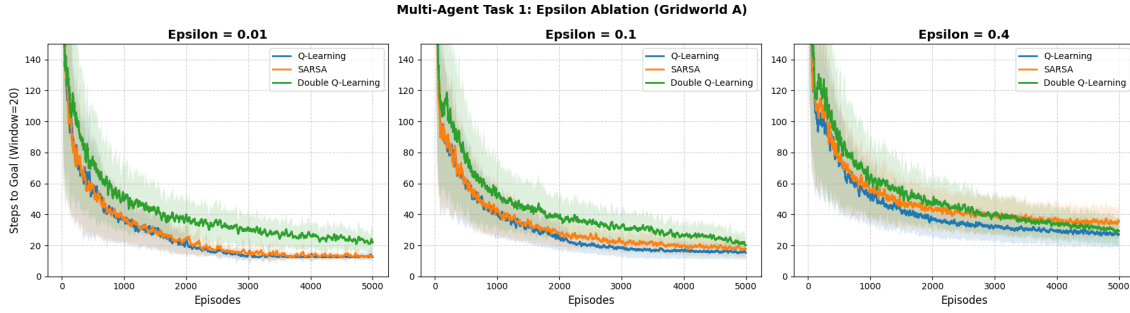


Figure 13: Epsilon hyperparameter ablation for Gridworld A, multi-agent case.

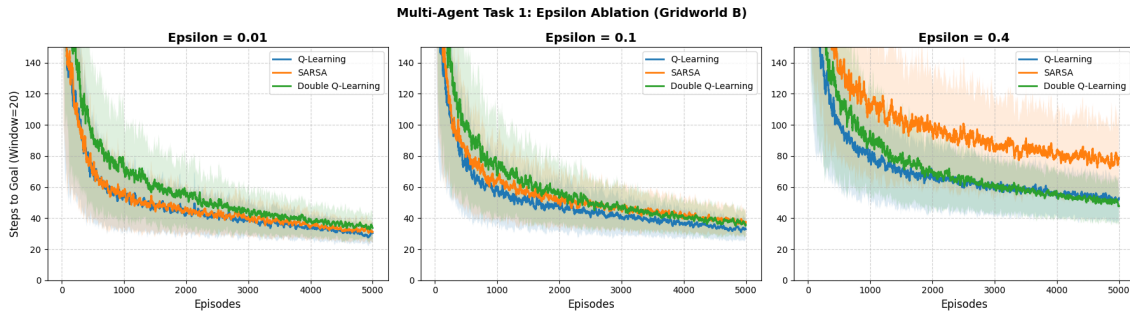


Figure 14: Epsilon hyperparameter ablation for Gridworld B, multi-agent case.

Gridworld A Figure 11, Figure 13

- **State Space Explosion:** Convergence requires thousands of episodes ($\sim 3,000-5,000$) compared to the single-agent's 100-200. This massive delay is caused by the joint state space expanding from 70 individual states to 4,900 combined states.
- **Collision Bottlenecks:** Baseline path efficiency is permanently inflated. Because the environment enforces physical blocking, agents create artificial traffic jams at the narrow gaps in the obstacle wall, forcing sub-optimal detours.
- **Learning Rate (α) Impact:** A low learning rate ($\alpha = 0.1$) is too sluggish for the massive state space, failing to fully plateau by 5,000 episodes. Aggressive updates ($\alpha = 0.9$) are strictly beneficial in this deterministic environment, allowing agents to map collision dynamics rapidly.
- **Epsilon (ϵ) Impact:** High exploration ($\epsilon = 0.4$) penalizes SARSA heavily. Because it is on-policy, SARSA learns to fear the narrow, high-collision bottlenecks, resulting in overly conservative and longer baseline paths compared to the off-policy agents.

Gridworld B Figure 12, Figure 14

- **Maximum Path Length Skew:** The overall step counts rest much higher (~ 30 -50 steps) than single-agent runs. Because the episode only terminates when the slowest agent reaches the goal, the graphs reflect the maximum path length of the two agents navigating the chaotic wind.
- **Off-Policy Robustness:** Q-Learning and Double Q-Learning remain remarkably stable even at $\alpha = 0.9$ and $\epsilon = 0.4$. Their target policies evaluate the optimal path, allowing them to ignore the noise of being blown away or physically blocked by the other agent.
- **On-Policy Fragility:** SARSA exhibits severe instability at $\alpha = 0.9$ and a catastrophic breakdown at $\epsilon = 0.4$ (plateauing near 80 steps). The frequent collision penalties, combined with wind shifts during forced exploration, cause SARSA to constantly unlearn its route and fail to converge efficiently.

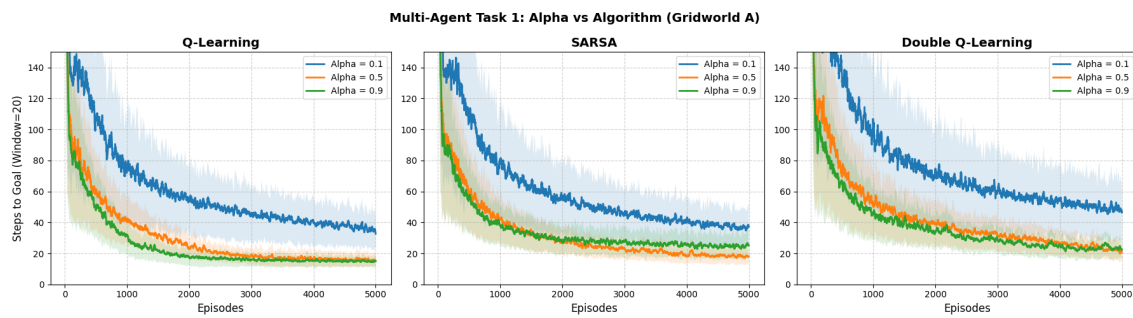


Figure 15: Alpha hyperparameter comparison, per algorithm, for Gridworld A, multi-agent case.

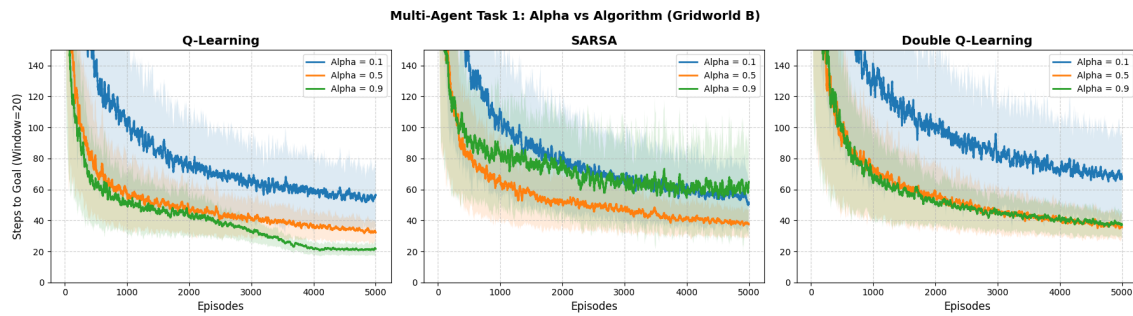


Figure 16: Alpha hyperparameter comparison, per algorithm, for Gridworld B, multi-agent case.

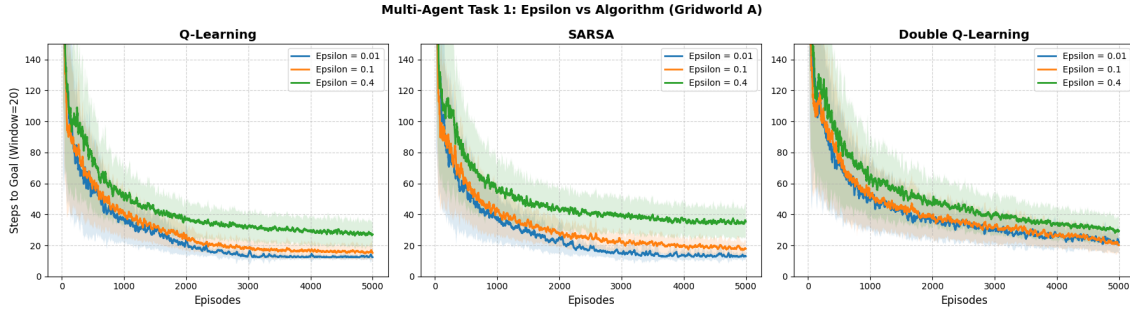


Figure 17: Epsilon hyperparameter comparison, per algorithm, for Gridworld A, multi-agent case.

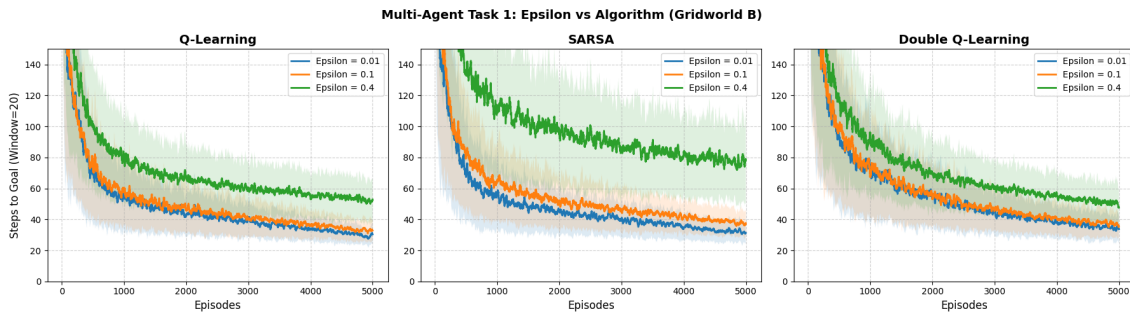


Figure 18: Epsilon hyperparameter comparison, per algorithm, for Gridworld B, multi-agent case.

Gridworld A Figure 15, Figure 17

- **Alpha:** Across all three algorithms, the learning rate (α) curves behave nearly identically. The massive 4,900-state joint space makes $\alpha = 0.1$ far too slow to converge within 5,000 episodes. Conversely, because the environment is deterministic, all algorithms fully trust new information, allowing $\alpha = 0.9$ to achieve the fastest and tightest convergence without instability.
- **Epsilon Disparity:** When forced to explore heavily ($\epsilon = 0.4$), the algorithms diverge. Q-Learning and Double Q-Learning establish a stable floor at ~ 25 -30 steps. SARSA, however, is on-policy; it anticipates the frequent collisions at the wall gaps and artificially downgrades those states, settling on a highly conservative, longer baseline of ~ 40 steps.

Gridworld B Figure 16, Figure 18

- **Off-Policy Superiority at High Alpha:** In Q-Learning and Double Q-Learning, the $\alpha = 0.9$ curve (green) represents the best performance, slicing through the noise of wind and collisions to rapidly map the shortest path.
- **SARSA's Alpha Breakdown:** Looking strictly at SARSA's alpha plot, the trend reverses: $\alpha = 0.9$ becomes the worst performer. When an agent collides and is subsequently swept away by the wind,

SARSA aggressively memorizes this penalty. A high learning rate causes it to permanently unlearn optimal routes, plateauing at a highly unstable ~ 60 steps.

- **Extreme Epsilon Penalties:** The wind strictly punishes the forced randomness of $\epsilon = 0.4$. While Q-Learning and Double Q-Learning manage to plateau around ~ 50 steps, SARSA's plot reveals a complete failure, resting at an inflated ~ 80 steps with massive variance bands.

5 Multi-Agent Task 2

We also ran task 2 with a multi-agent setup, with the results being shown in Figure 19 below:

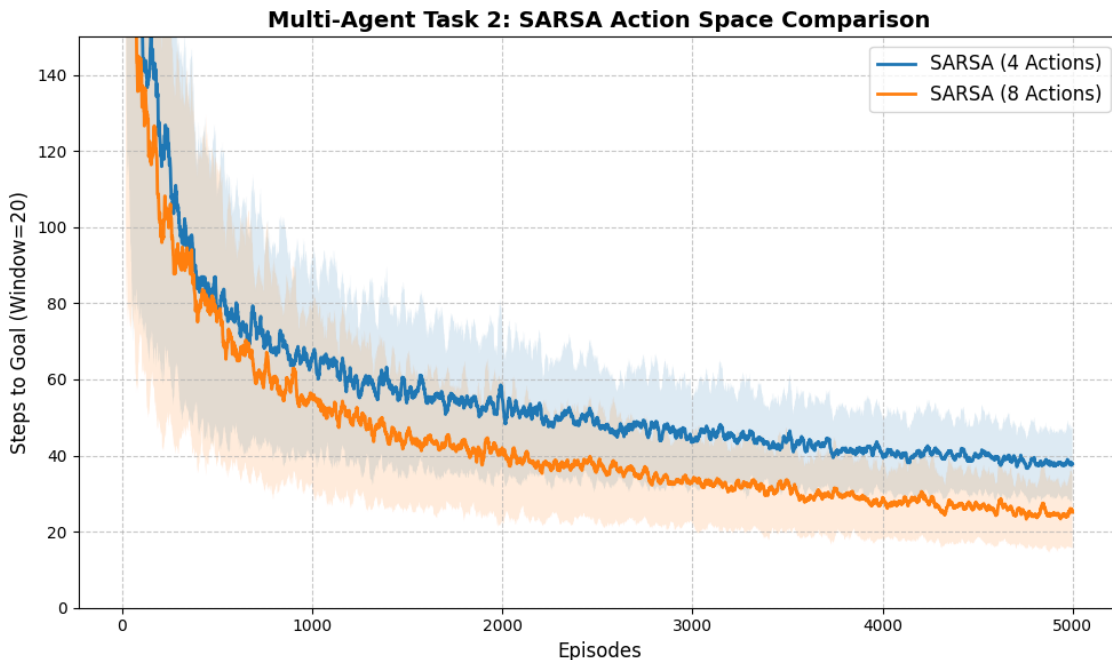


Figure 19: Multi-Agent SARSA Comparison on Girdworld B (4 moves vs. 8 moves).

- **Efficiency Gains:** Introducing diagonal moves dramatically improves path efficiency within the multi-agent environment. The 8-action model eventually plateaus at roughly ~ 25 steps per episode, whereas the 4-action model struggles to drop below ~ 40 steps.
- **Multi-Agent Penalty:** Crucially, even with the enhanced 8-action space, adding a second agent does not improve overall system efficiency; it strictly degrades it. The single-agent 8-action model reaches the goal in ~ 10 steps, while the multi-agent equivalent bottoms out at ~ 25 steps. The overhead of collision avoidance and waiting for the slower agent means navigating together is inherently less efficient than navigating alone.

- **Convergence Timelines:** Due to the massive 4,900-state joint space, both models require a vast amount of time to stabilize, continuing to trend downward even past 3,000 episodes, while the single-agent model converges to the optimum in 500 episodes